Evidence Report

Name:	Tytus Gonzalez
Title:	Capstone
Case:	26-T010
Date:	05/12/2026

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-05-11	0.1	Draft
Tytus Gonzalez	2026-05-12	1.0	Final

Executive Summary

On May 2026, a comprehensive digital forensic investigation was conducted on two forensic disk images identified as Device 1 and Device 2 as part of a forensic examination involving hidden data discovery, encrypted volume analysis, and deleted file recovery. The purpose of the investigation was to determine whether either device contained concealed, deleted, or encrypted information and to document the methods used to identify and recover potential evidence. All analysis was performed within a controlled forensic laboratory environment using a Linux-based forensic virtual machine to ensure evidence integrity and prevent modification of the original disk images.

The investigation began with an initial identification and verification phase in which both devices were examined to determine their partition layouts, file systems, and mounted volumes. Investigative procedures included the use of Linux forensic utilities such as lsblk, fdisk, and mount to enumerate connected storage devices and identify accessible partitions. Additional forensic analysis was performed using The Sleuth Kit (TSK) utilities, including commands such as mmls, fls, istat, and icat, to analyze file system structures, identify deleted artifacts, and examine metadata associated with files and directories. Manual hexadecimal analysis techniques were also incorporated throughout the investigation to inspect raw disk sectors and identify hidden data that was not visible through normal operating system navigation.

During the examination of Device 1, investigators determined that the mounted file systems alone did not contain all available data on the device. Further low-level disk analysis revealed hidden information stored outside of the mounted partitions within unallocated or non-file-system space. This hidden content was identified through direct hexadecimal examination of the disk image and analysis of sector offsets. Investigators documented the exact hexadecimal offset location where the concealed data was discovered and verified that the information was intentionally placed outside of standard file system structures to avoid detection during normal browsing operations. This portion of the analysis demonstrated how hidden data can exist independently of active partitions and highlighted the importance of examining raw disk structures in addition to mounted volumes.

Investigators also evaluated Device 1 for the presence of deleted files and recoverable artifacts. File system analysis techniques were used to inspect metadata entries and determine whether deleted file references remained within the file system tables. By analyzing inode and metadata structures using TSK utilities, investigators assessed whether deleted data could still be reconstructed or identified through residual file system information. The analysis documented the state of deleted artifacts and whether recoverable content remained accessible within unallocated space.

The examination of Device 2 involved the identification and analysis of an encrypted VeraCrypt volume. Investigators first identified the encrypted container and then performed forensic decryption procedures to gain access to its contents. Password recovery efforts and mounting procedures were conducted using VeraCrypt utilities within the forensic Linux environment. Once the encrypted volume was successfully mounted through the /dev/mapper/veracrypt1 device mapping, investigators gained access to the decrypted file system and analyzed its contents for evidence. Recovered files and artifacts from the decrypted volume were preserved by securely transferring them into the designated forensic evidence directory located within /mnt/Forensics/evidence.

Additional analysis was conducted on the decrypted VeraCrypt volume to determine whether deleted files or hidden artifacts existed within the encrypted file system. Investigators used TSK utilities and Linux file system analysis methods to inspect the mounted decrypted partition and evaluate whether deleted metadata entries or recoverable file remnants were present. The investigation also documented instances where mounted file systems could be identified successfully, but deleted files could not be recovered due to file system limitations, encryption characteristics, or the absence of recoverable metadata structures. These findings were important in distinguishing between accessible mounted partitions and the ability to successfully recover deleted content.

Throughout the investigation, investigators employed multiple forensic tactics to identify concealed information, including raw disk analysis, encrypted volume mounting, partition inspection, file system metadata analysis, and deleted file recovery techniques. The use of both automated forensic tools and manual analysis methods allowed investigators to validate findings across multiple layers of the disk images and ensure the accuracy of the examination results. Careful documentation was maintained throughout each stage of the investigation, including partition structures, mount points, hexadecimal offsets, recovered artifacts, and evidence transfer procedures.

The results of the investigation demonstrated several methods commonly used to conceal or protect digital evidence, including the use of VeraCrypt encryption, hidden data stored outside mounted partitions, and file deletion techniques designed to obscure artifacts from standard operating system visibility. By applying forensic recovery methodologies and low-level disk examination techniques, investigators successfully identified hidden information, mounted encrypted containers, and analyzed evidence that would not have been accessible through conventional file browsing methods. The investigation reinforced the importance of forensic methodology, detailed documentation, and specialized forensic tools when conducting examinations involving encrypted storage devices, deleted data, and hidden disk artifacts.

Synopsis

The Professor provided the following questions.

<u>DEVICE 1 & 2</u>

1. Is this disk partitioned? If so, with what scheme? (MBR or GPT)

2. How many partitions reside on this disk? (Hint: look at the slots section for entries with 000, etc. Not — or Meta)

<u>For each partition (or volume)</u>

1. What is the starting sector and file system associated with the partition?

2. What is the total size (in bytes) for the partition?

3. What is the volume serial number?

4. What is the volume label (name or ID), "None" if no entry?

5. What is the data structure used to track directory or file metadata for this file system? (i.e. FAT Directory Entries, MFT, Inodes)?

6. Does this partition/volume use encryption?

- **a. If so, what implementation is used?**
- **b. If so, what encryption algorithm was used?**
- **c. If so, what is the password to access the contents?**

7. Were there any deleted files on this image?

- **a. If so, what is the full path of the file deleted?**
- **b. If so, based on the extension of the file, what is it?**

8. Are there any credit card numbers on this device?

- **a. If so, what is the name and full path to the file containing credit card numbers?**
- **b. If so, what are the contents of this file?**

9. Does this volume contain any additional evidence supporting the claims of credit card theft?

10. What do you think this device was used for?

<u>BONUS</u>

1. Is there any hidden data on device 1 that is not visible in any of the mounted partitions? If so, what does it say and where did you find it? Give the exact offset of the data in hexadecimal output.

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
is4483-capstone-dev1.img	is4483-capsone-dev1	Image file of USB device provided by instructor
is4483-capstone-dev2.img	is4483-capstone-dev2	Image file of USB device provided by instructor

This section provides details of the digital evidence collected

Name	is4483-capstone-dev1.img
Type:	DOS/MBR boot sector; partition 1 : ID=0xee, start-CHS (0x0,0,2), end-CHS (0x3ff,255,63), startsector 1, 134217727 sectors,extended partition table (last)
Size:	68719476736 bytes
MD5:	DF59681778F78AB2DB0DD8DE8E9BEECC
SHA1:	8547E1E43DF12A0550CFC278371618FBBF207C08
SHA256	B65536C155EAD2C2F7976A0432DD6025DA998DF5BBB791F3F7D1C039E7327956

Name	is4483-capstone-dev2.img
Type:	DOS executable (COM, 0x8C-variant)
Size:	68719476736 bytes
MD5:	FE674640256F452E3493077C064F8E44
SHA1:	444C72F2A8335AC787B4F7F148A4B584D1417A0C
SHA256	D5B68BB302F09BBA515AD9E2CD2008FF5DE35FB9781C2A42028CEF26D8309B9C

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical / Virtual	Built
Windows 11x64_DFIR	Windows 11 Home	2021	Virtual	02/06/2026

Hardware

Name	Purpose	Additional Information
TdogExtreme	SanDisk Extreme Portable SSD	1TB

Software

Name	Version	Release	Purpose
The Sleuth Kit	4.14.0	April 14, 2025	A forensic toolset used to analyze disk images, recover files, and examine file system metadata in digital investigations.
VeraCrypt	1.26.24	May 30, 2025	A free open-source utility designed for on-the-fly encryption of data storage devices, volumes, and entire operating systems.

Case Information & Observables

This case involved analyzing 2 images from USB devices provided by the search warrant from law enforcement. The user of the images was Christopher Paul Bacon who is accused of stealing financial information. The following are my observations and responses to Professor's questions.

Observables

<u>DEVICE 1 & 2</u>

1. Is this disk partitioned? If so, with what scheme? (MBR or GPT)

For Device 1: Yes, the disk is partitioned using the GPT scheme. Analysis of the boot sector using 'mmls' revealed a GPT Header and a Partition Table which means it is a GPT-partitioned disk.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev1.img                07:53:23+0000
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0062916607   0062914560   primary
005:  001       0062916608   0125831167   0062914560   primary
006:  -------   0125831168   0134217727   0008386560   Unallocated
```

Figure 1: Using TSK 'mmls' command to display partition layout on dev1

For Device 2: No recognizable partition structure was identified on the disk. Analysis using 'mmls' did not detect a valid partition table, and examination of the first 512 bytes showed high-entropy randomized data with no identifiable filesystem or partition signatures. This suggests the image was not a traditional partitioned disk image, but instead a standalone VeraCrypt-encrypted container or volume, that the image may contain an encrypted container, or the device was intentionally obfuscated.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev2.img


/mnt/Forensics ❯
```

Figure 2: Using TSK 'mmls' command to display partition layout on dev2

```
/mnt/Forensics ❯ xxd -l 512 is4483-capstone-dev2.img                  07:56:06
00000000: 8cda 372f c6e7 3778 4c05 913e 2fd3 6251  ..7/..7xL..>/.bQ
00000010: a06b 0671 189d 5c5f d884 0e6d 6ad0 24b4  .k.q..\_...mj.$.
00000020: 96a7 479b fb43 f700 b8ee a03d 6871 3611  ..G..C.....=hq6.
00000030: 8fb1 9370 0966 c6d8 92db 9dec 745c 43ff  ...p.f......t\C.
00000040: 03fb 0e3c 9fbd 8142 e098 2f67 ba4b eda4  ...<...B../g.K..
00000050: ea5f 3251 902d 79bb f06e dad0 941e 6e47  ._2Q.-y..n....nG
00000060: 4d4b 5ce9 814c 1ebe 98cb a9bd 51fd da9e  MK\..L......Q...
00000070: e6d9 8b6a 1693 23fd b44c ec2d 3590 0908  ...j..#..L.-5...
00000080: ce8f 2cf3 7bb1 9ac8 9dc0 8c4d 5e61 0cc7  ..,.{......M^a..
00000090: 6f8d 0b9a 4a5d 32bb 85ae f9eb 91ba de75  o...J]2........u
000000a0: 4e6a f767 aae0 33d2 fffd 082a dd43 9ac9  Nj.g..3....*.C..
000000b0: a42c 230b 0302 6465 9766 280d e318 dbd7  .,#...de.f(.....
000000c0: 00e3 5ce0 8200 181c 1e3d 2f6d b82c 7757  ..\......=/m.,wW
000000d0: d941 a1ac 033b d1cd 789a ea2e 9752 d5a5  .A...;..x....R..
000000e0: 1030 eb20 9c81 2630 58a1 db98 5a3b 39d8  .0. ..&0X...Z;9.
000000f0: 9bd5 c0c6 13de ebbe 6a4e ac51 bace 8377  ........jN.Q...w
00000100: a6da 905f 91a5 2f78 66f1 bffd 5377 bb98  ..._../xf...Sw..
00000110: 56d3 7889 6503 871b 2432 52da 68fe c2f2  V.x.e...$2R.h...
00000120: 696d 4837 7248 0ccc 4494 66c1 ed03 e0bc  imH7rH..D.f.....
00000130: 2fc2 c6be b998 0f56 3c75 90b3 85cd d26e  /......V<u.....n
00000140: 8bb8 7fbc 1222 ab25 717c 52e3 0cef 89f2  .....".%q|R.....
00000150: b6c5 7f44 c68d 63de 6388 a52e 6c6a c665  ...D..c.c...lj.e
00000160: f734 74b2 6b6f c1b1 de96 e101 e811 92ae  .4t.ko..........
00000170: 6dfb 0287 af23 8691 aae8 d7ec eab6 e8f6  m....#..........
00000180: 0afc 145d eadc dcb5 bed2 1646 0fdf 573e  ...].......F..W>
00000190: 32b7 25ad b711 e7be 3bd4 4ba8 dd00 a64f  2.%.....;.K....O
000001a0: 0e83 9bba 56c7 2915 0531 36e0 c21d 24fb  ....V.)..16...$.
000001b0: e2aa 0fc8 3e02 8cb9 d968 d4e8 17d8 9865  ....>....h.....e
000001c0: 294d 7f75 f681 7b73 4c0c 6f1e 6781 b695  )M.u..{sL.o.g...
000001d0: 32e9 f607 868a 0c01 decb d984 791e ce73  2...........y..s
000001e0: 503e 2524 de0b c8b0 297d 91bb 1ad7 13f8  P>%$....)}......
000001f0: 8ed5 4956 cb4c b588 3388 ebb1 6f79 4781  ..IV.L..3...oyG.
```

Figure 3: Using 'xxd' command to create hexadecimal dump of first 512 bytes of dev2

```
/mnt/Forensics ❯ fsstat is4483-capstone-dev2.img
Possible encryption detected (High entropy (8.00))
```

Figure 4: Using TSK 'fsstat' command to show status of file system of dev2

2. How many partitions reside on this disk? (Hint: look at the slots section for entries with 000, etc. Not — or Meta)

For Device 1: Device 1 contains two primary partitions within a GPT structure. Analysis with ‘mmls’ identified two valid partition entries (000 and 001). Additional entries labeled as metadata, GPT header, partition table, and unallocated space were not counted as partitions.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev1.img                          07:53:23+0000
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0062916607   0062914560   primary
005:  001       0062916608   0125831167   0062914560   primary
006:  -------   0125831168   0134217727   0008386560   Unallocated
```

Figure 5: Using TSK 'mmls' command to display partition layout on dev1

For Device 2: No recognizable partitions were identified on Device 2.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev2.img


/mnt/Forensics ❯
```

Figure 6: Using TSK 'mmls' command to display partition layout on dev2

<u>For each partition (or volume)</u>

1. What is the starting sector and file system associated with the partition?

For Device 1: The file system associated with the 2 partitions in dev1 were NTFS and FAT32. The starting sector for each partition was found using 'mmls', and Partition 000 started at 2048, and Partition 001 started at 62916608. Partition 000's file system was NTFS and Partition 001's file system was FAT32.

```
/mnt/Forensics ❯ fsstat -o 2048 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 09B25B8B3B0241BE
OEM Name: NTFS
Volume Name: PERSONAL
Version: Windows XP
```

Figure 7: Using TSK 'fsstat' command to show detailed structure of partition 1 of dev1

```
/mnt/Forensics ❯ fsstat -o 62916608 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0xd4504451
Volume Label (Boot Sector): DATACYTE
Volume Label (Root Directory): DATACYTE
File System Type Label: FAT32
Next Free Sector (FS Info): 30912
Free Sector Count (FS Info): 62883584
```

Figure 8: Using TSK 'fsstat' command to show detailed structure of partition 2 of dev1

For Device 2: Before decryption, the filesystem on Device 2 could not be identified using traditional forensic tools because the image was stored as a VeraCrypt encrypted container. Attempts to analyze the image directly with 'mmls' and 'fsstat' were unsuccessful due to the encrypted volume. After successfully decrypting and mounting the container using VeraCrypt, the file system was identified as FAT32 (vfat), but there is no starting sector identified.

```
/mnt/Forensics ❯ fsstat is4483-capstone-dev2.img
Possible encryption detected (High entropy (8.00))
```

Figure 9: Using TSK 'fsstat' command to show status of file system of dev2

```
/mnt/Forensics ❯ lsblk -f
NAME          FSTYPE FSVER LABEL  UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0
└─veracrypt1 vfat   FAT32 SECURE D7E4-78CA
zram0         swap   1     zram0  14c93ba3-7fe3-400f-aeb4-5703cb648502                [SWAP]
nvme0n1
├─nvme0n1p1
├─nvme0n1p2  ext4   1.0          e08e906c-abab-4a69-a11a-fc671ca2d40b  496.3M    42% /boot
└─nvme0n1p3  btrfs        fedora 0f9b4d6a-2164-41bc-9586-f3c6288c51e4    128G    12% /home
                                                                                      /

/mnt/Forensics ❯
```

Figure 10: Using 'lsblk' command to show file system information of dev2 after decryption

2. What is the total size (in bytes) for the partition?

For Device 1: Analysis with 'mmls' determined that both partitions on Device 1 contain 62,914,560 sectors. Using 512-byte sectors, each partition has a total size of 32,212,254,720 bytes. This is calculated by 62,914,560 * 512 = 32,212,254,720.

```
/mnt/Forensics › mmls is4483-capstone-dev1.img                07:53:23+0000
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0062916607   0062914560   primary
005:  001       0062916608   0125831167   0062914560   primary
006:  -------   0125831168   0134217727   0008386560   Unallocated
```

Figure 11: Using TSK 'mmls' command to display partition layout on dev1

For Device 2: Unable to determine prior to decryption because no accessible partition structure metadata was identified.

3. What is the volume serial number?

For Device 1: The volume serial numbers of Device 1 were identified using ‘fsstat’. The volume serial number of Partition 000 was 09B25B8B3B0241BE, and the volume serial number of Partition 001 was 0xd4504451.

```
/mnt/Forensics › fsstat -o 2048 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 09B25B8B3B0241BE
OEM Name: NTFS
Volume Name: PERSONAL
Version: Windows XP
```

Figure 12: Using TSK 'fsstat' command to show detailed structure of partition 1 of dev1

```
/mnt/Forensics ❯ fsstat -o 62916608 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0xd4504451
Volume Label (Boot Sector): DATACYTE
Volume Label (Root Directory): DATACYTE
File System Type Label: FAT32
Next Free Sector (FS Info): 30912
Free Sector Count (FS Info): 62883584
```

Figure 13: Using TSK 'fsstat' command to show detailed structure of partition 2 of dev1

For Device 2: Before decryption, the filesystem on Device 2 could not be identified using traditional forensic tools because the image was stored as a VeraCrypt encrypted container. Attempts to analyze the image directly with 'mmls' and 'fsstat' were unsuccessful due to the encrypted volume. After successfully decrypting and mounting the container using VeraCrypt, the volume serial label was identified as D7E4-78CA using 'lsblk' command.

4. What is the volume label (name or ID), "None" if no entry?

For Device 1: The volume labels of Device 1 were identified using 'fsstat'. The volume label of Partition 000 was PERSONAL, and the volume serial number of Partition 001 was DATACYTE.

```
/mnt/Forensics ❯ fsstat -o 2048 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: 09B25B8B3B0241BE
OEM Name: NTFS
Volume Name: PERSONAL
Version: Windows XP
```

Figure 14: Using TSK 'fsstat' command to show detailed structure of partition 1 of dev1

```
/mnt/Forensics ❯ fsstat -o 62916608 is4483-capstone-dev1.img
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT32

OEM Name: mkfs.fat
Volume ID: 0xd4504451
Volume Label (Boot Sector): DATACYTE
Volume Label (Root Directory): DATACYTE
File System Type Label: FAT32
Next Free Sector (FS Info): 30912
Free Sector Count (FS Info): 62883584
```

Figure 15: Using TSK 'fsstat' command to show detailed structure of partition 2 of dev1

For Device 2: Before decryption, the filesystem on Device 2 could not be identified using traditional forensic tools because the image was stored as a VeraCrypt encrypted container. Attempts to analyze the image directly with 'mmls' and 'fsstat' were unsuccessful due to the encrypted volume. After successfully decrypting and mounting the container using VeraCrypt, the volume label was identified as "SECURE".

```
/mnt/Forensics ❯ lsblk -f
NAME         FSTYPE FSVER LABEL  UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0
└─veracrypt1 vfat   FAT32 SECURE D7E4-78CA
zram0        swap   1     zram0  14c93ba3-7fe3-400f-aeb4-5703cb648502                [SWAP]
nvme0n1
├─nvme0n1p1
├─nvme0n1p2  ext4   1.0          e08e906c-abab-4a69-a11a-fc671ca2d40b  496.3M    42% /boot
└─nvme0n1p3  btrfs        fedora 0f9b4d6a-2164-41bc-9586-f3c6288c51e4    128G    12% /home
                                                                                     /

/mnt/Forensics ❯
```

Figure 16: Using 'lsblk' command to show file system information of dev2 after decryption

5. What is the data structure used to track directory or file metadata for this file system? (i.e. FAT Directory Entries, MFT, Inodes)?

For Device 1: Partition 000 (NTFS) uses the MFT to track file and directory metadata. Partition 001 (FAT32) uses FAT directory entries to track metadata. This is standard for these file systems.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev1.img                         07:53:23+0000
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0062916607   0062914560   primary
005:  001       0062916608   0125831167   0062914560   primary
006:  -------   0125831168   0134217727   0008386560   Unallocated
```

Figure 17: Using TSK 'mmls' command to display partition layout on dev1

For Device 2: After decrypting and mounting the VeraCrypt volume, the filesystem was identified as FAT32 (vfat). FAT32 uses directory entries to track file and directory metadata.

6. Does this partition/volume use encryption?

For Device 1: No evidence of encryption was identified on Device 1 during the initial file system analysis. A hexadecimal dump was taken of the first 512 bytes and there were no signs of encryption. Also, the device partition table was correctly output using ‘mmls’.

For Device 2: Yes, there is likely encryption on Device 2 based on the ‘fsstat’ command showing there was possible encryption detection.

a. If so, what implementation is used?

For Device 1: N/A

For Device 2: The encryption implementation used on the volume was VeraCrypt.

b. If so, what encryption algorithm was used?

For Device 1: N/A

For Device 2: The volume utilized VeraCrypt encryption. The exact algorithm was found to be AES operating in XTS mode with a 256-bit primary key and 256-bit secondary key.

c. If so, what is the password to access the contents?

For Device 1: N/A

For Device 2: The recovered password used to access the encrypted volume was *supersecret123*.

```
/mnt/Forensics ❯ mmls is4483-capstone-dev1.img                          07:53:23+0000
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0062916607   0062914560   primary
005:  001       0062916608   0125831167   0062914560   primary
006:  -------   0125831168   0134217727   0008386560   Unallocated
```

Figure 18: Using TSK 'mmls' command to display partition layout on dev1

```
/mnt/Forensics ❯ xxd -l 512 is4483-capstone-dev1.img
00000000: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000040: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000180: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000190: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001c0: 0200 eeff ffff 0100 0000 ffff ff07 0000  ................
000001d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001f0: 0000 0000 0000 0000 0000 0000 0000 55aa  ..............U.
```

Figure 19: Using 'xxd' command to create hexadecimal dump of first 512 bytes of dev 1

```
/mnt/Forensics ❯ fsstat is4483-capstone-dev2.img
Possible encryption detected (High entropy (8.00))
```

Figure 20: Using TSK 'fsstat' command to show status of file system of dev2

```
~ › sudo veracrypt --text --mount /mnt/Forensics/is4483-capstone-dev2.img /mnt/Forensics/evidence --password "supersecret123"
--pim 0 --keyfiles "" --non-interactive
[sudo] password for hatter:

took 8s
~ › ls/mnt/Forensics/evidence                                                                              10:18:53+0000
fish: ls/mnt/Forensics/evidence: command not found...


~ › ls /mnt/Forensics/evidence/                                                                            10:19:11+0000
Permissions Size User   Group  Date Modified    Name
.rwx------@  206 hatter hatter 2025-05-05 20:11 bank_accounts.csv
.rwx------@  235 hatter hatter 2025-05-05 20:11 buyers_list.txt
.rwx------@  311 hatter hatter 2025-05-05 20:11 cc_list.txt
.rwx------@   19 hatter hatter 2025-05-05 20:11 keylogger.exe
.rwx------@   19 hatter hatter 2025-05-05 20:11 ransomware.exe
.rwx------@   19 hatter hatter 2025-05-05 20:11 stealer.exe
```

Figure 21: Mounting evidence using VeraCrypt and the password to browse the decrypted contents of dev2

```
/mnt/Forensics/evidence › sudo veracrypt --text --volume-properties /mnt/Forensics/is4483-capstone-dev2.img      19:23:54+0000
[sudo] password for hatter:
shell-init: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
job-working-directory: error retrieving current directory: getcwd: cannot access parent directories: No such file or directory
Slot: 1
Volume: /mnt/Forensics/is4483-capstone-dev2.img
Virtual Device: /dev/mapper/veracrypt1
Mount Directory:
Size: 64.0 GiB
Type: Normal
Read-Only: No
Hidden Volume Protected: No
Encryption Algorithm: AES
Primary Key Size: 256 bits
Secondary Key Size (XTS Mode): 256 bits
Block Size: 128 bits
Mode of Operation: XTS
PKCS-5 PRF: HMAC-SHA-512
Volume Format Version: 2
Embedded Backup Header: Yes
```

Figure 22: Running 'veracrypt' command to find properties of the evidence of dev2

7. Were there any deleted files on this image?

For Device 1: Yes, deleted files were found on this image using the TSK 'fls' identified multiple deleted files within the image

For Device 2: Deleted-file analysis was attempted on the decrypted VeraCrypt volume using TSK 'fls'. However, the filesystem could not be interpreted directly through the VeraCrypt mapper device, preventing reliable recovery of deleted entries.

a. If so, what is the full path of the file deleted?

For Device 1: One deleted file identified during analysis was: *TechnicalSpecs/cc_list.txt* (found on Partition 000). Additional deleted orphan files were also identified within the *$OrphanFiles* directory structure (found on Partition 001).

For Device 2: No recoverable deleted file paths were identified during analysis of Device 2.

b. If so, based on the extension of the file, what is it?

For Device 1: The deleted files *cc_list.txt* was a text file (.txt) and appeared to contain credit card related information based on the filename of the image.

For Device 2: N/A

```
/mnt/Forensics ❯ fls -r -d -p -o 2048 /mnt/Forensics/is4483-capstone-dev1.img
-/r * 68-128-2: Pictures/family3.jpg
-/r * 16:       $OrphanFiles/OrphanFile-16
-/r * 17:       $OrphanFiles/OrphanFile-17
-/r * 18:       $OrphanFiles/OrphanFile-18
-/r * 19:       $OrphanFiles/OrphanFile-19
-/r * 20:       $OrphanFiles/OrphanFile-20
-/r * 21:       $OrphanFiles/OrphanFile-21
-/r * 22:       $OrphanFiles/OrphanFile-22
-/r * 23:       $OrphanFiles/OrphanFile-23


/mnt/Forensics ❯ fls -r -d -p -o 62916608 /mnt/Forensics/is4483-capstone-dev1.img
r/r * 518:      TechnicalSpecs/cc_list.txt
```

Figure 23: Using TSK 'fls' command to do a recursive search for any deleted files on dev1

```
/mnt/Forensics ❯ sudo fls -r -d -p /dev/mapper/veracrypt1
[sudo] password for hatter:
Cannot determine file system type (raw_read: file "/dev/mapper/veracrypt1" offset: 0 read len: 1024 - Input/output error)

took 4s
/mnt/Forensics ❯ sudo fsstat /dev/mapper/veracrypt1
[sudo] password for hatter:
Cannot determine file system type (raw_read: file "/dev/mapper/veracrypt1" offset: 0 read len: 1024 - Input/output error)
```

Figure 24: Using TSK 'fls' and 'fsstat' command to look for files or metadata about the decrypted block from dev2

8. Are there any credit card numbers on this device?

For Device 1: Yes, the volume contained a file called *cc_list.txtI* which contained

a record of people's names and credit card numbers. This suggests malicious activity pertaining to the user Christopher P. Bacon because this is likely stolen information.

For Device 2: Yes, the mounted encrypted volume contained files associated with financial data and potential credit card information.

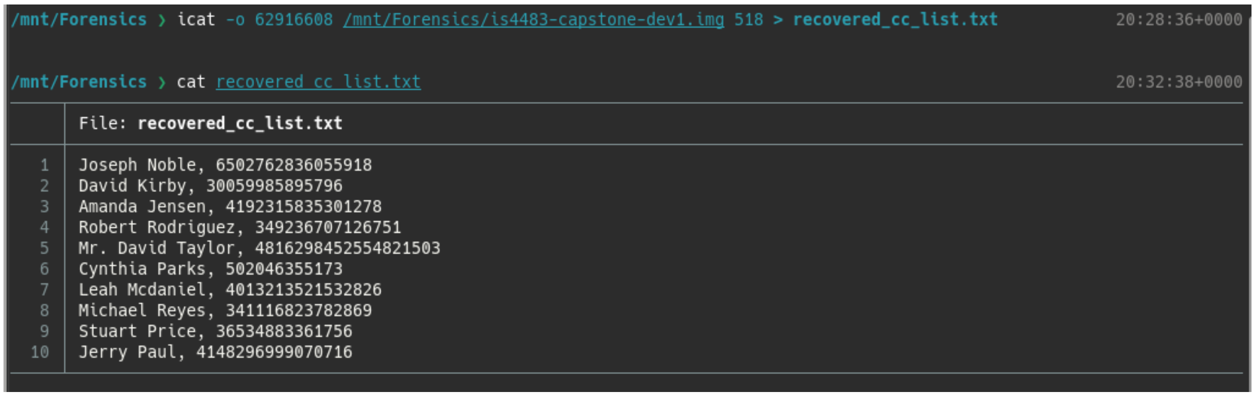
a. If so, what is the name and full path to the file containing credit card numbers?

For Device 1: The file identified as containing credit card information was *TechnicalSpecs/cc_list.txt*. The file was identified during deleted-file analysis using TSK 'fls' and recovered using 'icat'.

For Device 2: The file identified as containing credit card information was */mnt/Forensics/evidence/cc_list.txt.* The file was recovered after successfully decrypting and mounting the VeraCrypt volume.

b. If so, what are the contents of this file?

For Device 1: The recovered text file contained multiple names paired with credit card numbers. The contents strongly suggest that the file was used to store stolen or unauthorized payment card information.

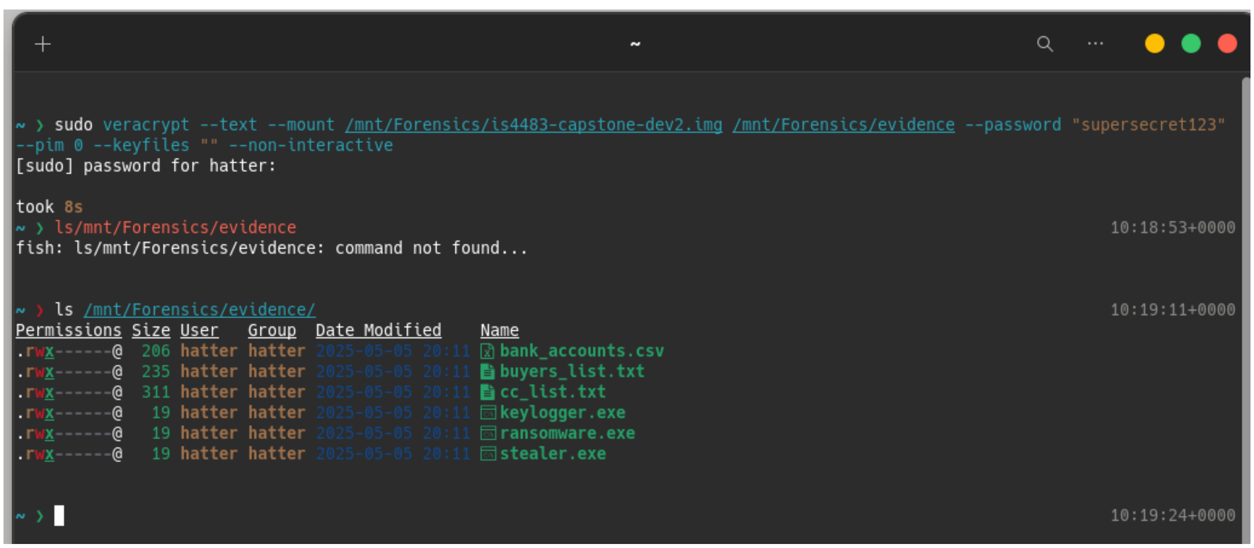


```
/mnt/Forensics ❯ icat -o 62916608 /mnt/Forensics/is4483-capstone-dev1.img 518 > recovered_cc_list.txt     20:28:36+0000

/mnt/Forensics ❯ cat recovered_cc_list.txt                                                                20:32:38+0000
     File: recovered_cc_list.txt
  1  Joseph Noble, 6502762836055918
  2  David Kirby, 30059985895796
  3  Amanda Jensen, 4192315835301278
  4  Robert Rodriguez, 349236707126751
  5  Mr. David Taylor, 4816298452554821503
  6  Cynthia Parks, 502046355173
  7  Leah Mcdaniel, 4013213521532826
  8  Michael Reyes, 341116823782869
  9  Stuart Price, 36534883361756
 10  Jerry Paul, 4148296999070716
```

Figure 25: Using TSK 'icat' command to recover 'cc_list.txt' and view the contents from dev 1

For Device 2: The text file contained multiple names paired with credit card numbers. The contents of the file indicate the storage of sensitive financial information consistent with stolen or illicitly obtained payment card data.



```
~ ❯ sudo veracrypt --text --mount /mnt/Forensics/is4483-capstone-dev2.img /mnt/Forensics/evidence --password "supersecret123"
--pim 0 --keyfiles "" --non-interactive
[sudo] password for hatter:

took 8s
~ ❯ ls/mnt/Forensics/evidence                                                                10:18:53+0000
fish: ls/mnt/Forensics/evidence: command not found...

~ ❯ ls /mnt/Forensics/evidence/                                                              10:19:11+0000
Permissions Size User   Group  Date Modified    Name
.rwx------@  206 hatter hatter 2025-05-05 20:11 bank_accounts.csv
.rwx------@  235 hatter hatter 2025-05-05 20:11 buyers_list.txt
.rwx------@  311 hatter hatter 2025-05-05 20:11 cc_list.txt
.rwx------@   19 hatter hatter 2025-05-05 20:11 keylogger.exe
.rwx------@   19 hatter hatter 2025-05-05 20:11 ransomware.exe
.rwx------@   19 hatter hatter 2025-05-05 20:11 stealer.exe

~ ❯                                                                                          10:19:24+0000
```

Figure 26: Mounting evidence using VeraCrypt and the password to browse the decrypted contents of dev2

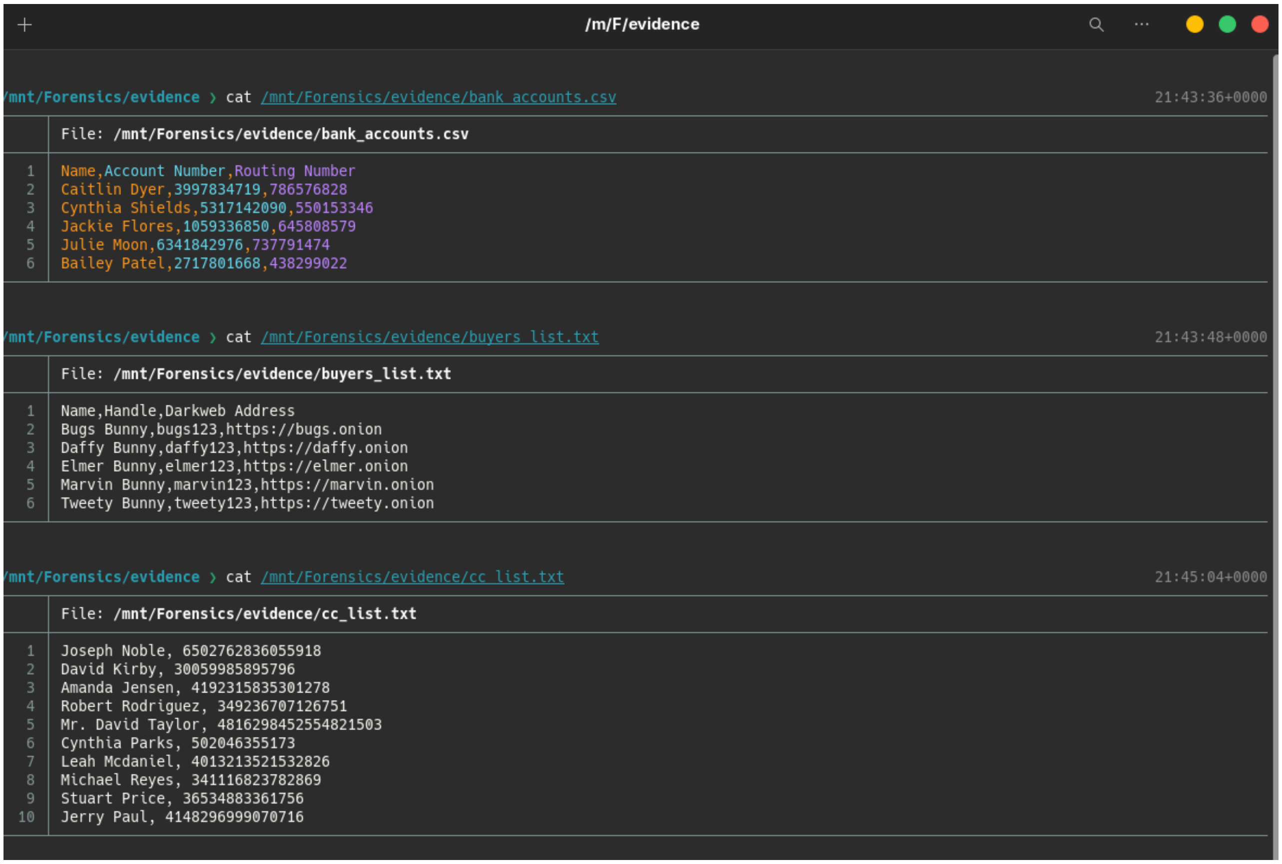


Figure 27: Using 'cat' command to display the content of the suspected files that may include sensitive bank/credit card information on dev2

9. Does this volume contain any additional evidence supporting the claims of credit card theft?

For Device 1: Yes, additional evidence recovered from Device 1 supports the conclusion that the system was used to store and conceal stolen financial information. Deleted-file analysis identified a deleted text file named *TechnicalSpecs/cc_list.txt*. The filename suggests the file contained credit card information. The fact that the file was deleted and multiple orphaned files were deleted indicates a possible attempt to conceal evidence.

```
/mnt/Forensics ❯ fls -r -d -p -o 2048 /mnt/Forensics/is4483-capstone-dev1.img
-/r * 68-128-2: Pictures/family3.jpg
-/r * 16:       $OrphanFiles/OrphanFile-16
-/r * 17:       $OrphanFiles/OrphanFile-17
-/r * 18:       $OrphanFiles/OrphanFile-18
-/r * 19:       $OrphanFiles/OrphanFile-19
-/r * 20:       $OrphanFiles/OrphanFile-20
-/r * 21:       $OrphanFiles/OrphanFile-21
-/r * 22:       $OrphanFiles/OrphanFile-22
-/r * 23:       $OrphanFiles/OrphanFile-23


/mnt/Forensics ❯ fls -r -d -p -o 62916608 /mnt/Forensics/is4483-capstone-dev1.img
r/r * 518:      TechnicalSpecs/cc_list.txt
```

Figure 28: Using TSK 'fls' command to do a recursive search for any deleted files on dev1

```
/mnt/Forensics ❯ icat -o 62916608 /mnt/Forensics/is4483-capstone-dev1.img 518 > recovered_cc_list.txt    20:28:36+0000

/mnt/Forensics ❯ cat recovered_cc_list.txt                                                                20:32:38+0000
File: recovered_cc_list.txt
Joseph Noble, 6502762836055918
David Kirby, 30059985895796
Amanda Jensen, 4192315835301278
Robert Rodriguez, 349236707126751
Mr. David Taylor, 4816298452554821503
Cynthia Parks, 502046355173
Leah Mcdaniel, 4013213521532826
Michael Reyes, 341116823782869
Stuart Price, 36534883361756
Jerry Paul, 4148296999070716
```

Figure 29: Using TSK 'icat' command to recover 'cc_list.txt' and view the contents from dev 1

For Device 2: Yes, significant additional evidence supports the claim that credit card data was stolen. The mounted filesystem contained several suspicious files directly associated with financial theft, and criminal activity. Analysis of *bank_accounts.csv* revealed multiple names paired with bank account and routing numbers. The *buyers_list.txt* contained usernames and associated dark web '.onion' addresses, suggesting communication to illicit buyers. The *cc_list.txt* file contained multiple names paired with payment card numbers consistent with stolen credit card data. Also, the presence of malicious executables such as keylogger.exe, stealer.exe, and ransomware.exe strongly suggest that the device was used to facilitate credential theft and data exfiltration related to financial fraud.

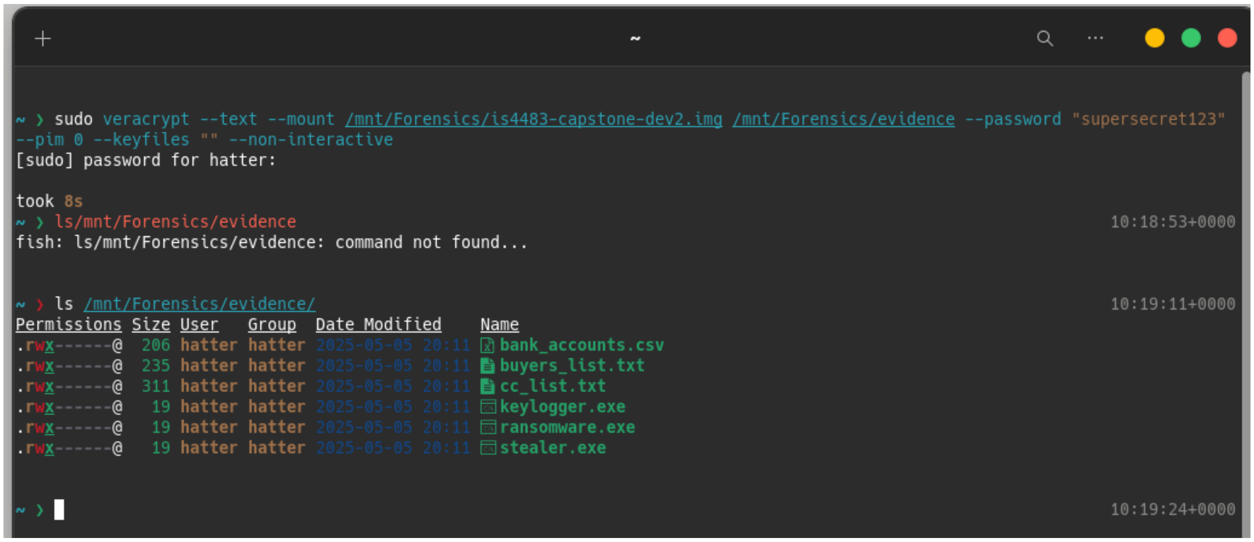


Figure 30: Mounting evidence using VeraCrypt and the password to browse the decrypted contents of dev2

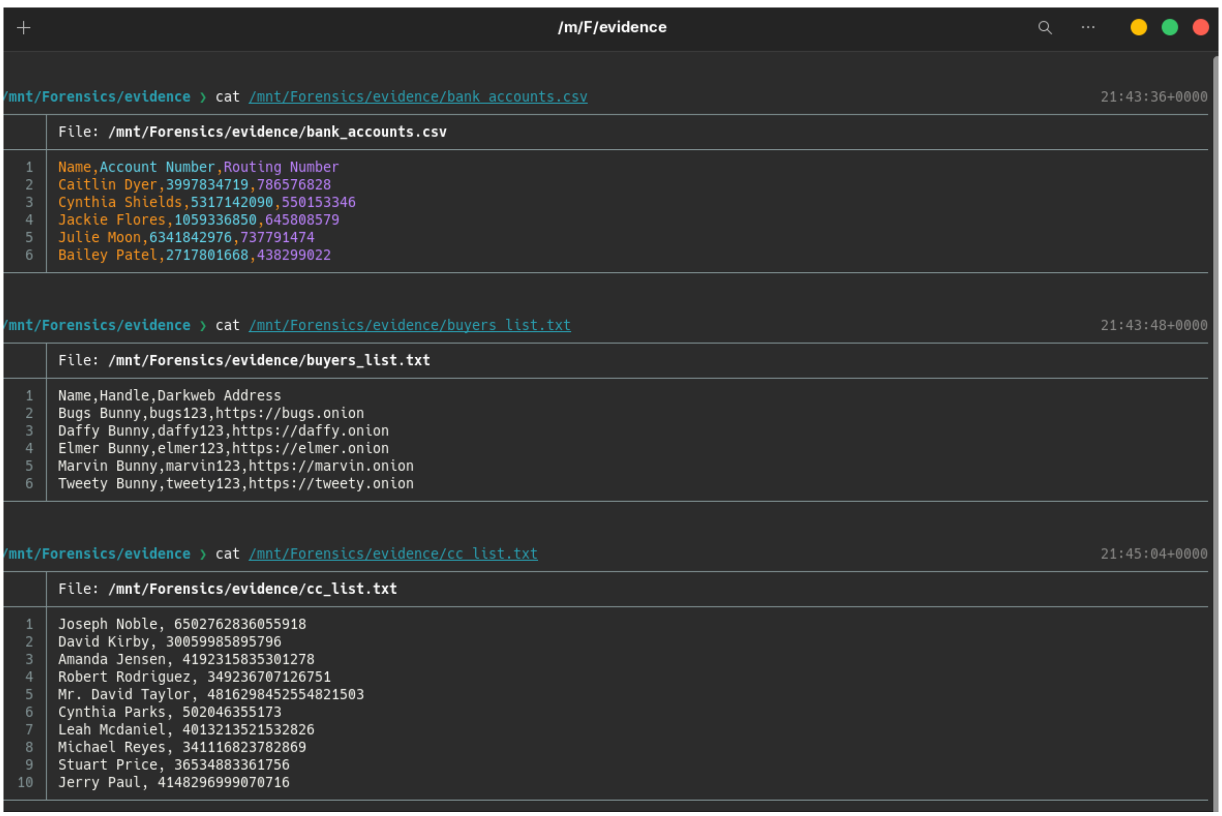


Figure 31: Using 'cat' command to display the content of the suspected files that may include sensitive bank/credit card information on dev2

10. What do you think this device was used for?

For Device 1: Based on forensic evidence, Device 1 appears to have been used to store and manage stolen financial information. The recovery of a deleted file associated with credit card data suggests the device contained sensitive financial records that were intentionally deleted to conceal evidence. The presence of deleted and orphaned files indicates attempts to remove or obscure potentially incriminating information from the system.

For Device 2: Based on forensic evidence, Device 2 appears to have been used as a storage location for stolen financial information and malicious cybercrime tools. The use of VeraCrypt AES encryption shows an intentional attempt to hide the contents of the volume. The decrypted volume contained stolen credit card numbers, bank accounts, dark web buyer information, and multiple malicious executables. These findings suggest that the device was used in support of cybercriminal operations involving credential theft, financial fraud, storing stolen payment information, and potential intent to sell data through the dark web.

BONUS

1. Is there any hidden data on device 1 that is not visible in any of the mounted partitions? If so, what does it say and where did you find it? Give the exact offset of the data in hexadecimal output.

Yes, hidden residual data was identified on Device 1 within unallocated space that was not visible through the mounted partitions. Analysis using 'blkls' recovered deleted credit card information associated with the previously deleted *cc_list.txt* file. The hidden data was recovered from unallocated space using TSK and confirmed within the original disk image at hexadecimal offset 0x222. Using the grep command, I was able to find the decimal offset of the start of the hidden data.

```
~
:...skipping...
Joseph Noble, 6502762836055918
David Kirby, 30059985895796
Amanda Jensen, 4192315835301278
Robert Rodriguez, 349236707126751
Mr. David Taylor, 4816298452554821503
Cynthia Parks, 502046355173
Leah Mcdaniel, 4013213521532826
Michael Reyes, 341116823782869
Stuart Price, 36534883361756
Jerry Paul, 4148296999070716
~
(END)
```

```
/mnt/hgfs/Forensics ❯ grep -oba "Joseph Noble, 6502762836055918" /mnt/hgfs/Forensics/is4483-capstone-dev1.img
546:32229081088:Joseph Noble, 6502762836055918
printf '0x%X\n' 546

took 3m37s
/mnt/hgfs/Forensics ❯ printf '0x%X\n' 546
0x222


/mnt/hgfs/Forensics ❯
```

Conclusion

This assignment provided valuable hands-on experience with real-world digital forensic investigation techniques, including partition analysis, deleted-file recovery, encryption analysis, and evidence examination using tools such as The Sleuth Kit and VeraCrypt. One of the most challenging aspects of the assignment was working with the encrypted VeraCrypt container and understanding the difference between analyzing a traditional disk image and an encrypted volume. Troubleshooting filesystem access and mounting issues helped strengthen my understanding of forensic workflows and how encryption can complicate investigations. Overall, this assignment was effective in demonstrating how multiple forensic tools and techniques can be combined to recover, analyze, and interpret digital evidence in a realistic investigation scenario.

Exhibit A – Stored Banking/Credit Card Information from Devices

```
/mnt/Forensics ❯ icat -o 62916608 /mnt/Forensics/is4483-capstone-dev1.img 518 > recovered_cc_list.txt          20:28:36+0000

/mnt/Forensics ❯ cat recovered_cc_list.txt                                                                        20:32:38+0000
       File: recovered_cc_list.txt

       Joseph Noble, 6502762836055918
       David Kirby, 30059985895796
       Amanda Jensen, 4192315835301278
       Robert Rodriguez, 349236707126751
       Mr. David Taylor, 4816298452554821503
       Cynthia Parks, 502046355173
       Leah Mcdaniel, 4013213521532826
       Michael Reyes, 341116823782869
       Stuart Price, 36534883361756
       Jerry Paul, 4148296999070716
```

Figure 32: Device 1 'cc_list.txt'

```
/m/F/evidence

/mnt/Forensics/evidence ❯ cat /mnt/Forensics/evidence/bank_accounts.csv                                           21:43:36+0000
       File: /mnt/Forensics/evidence/bank_accounts.csv

       Name,Account Number,Routing Number
       Caitlin Dyer,3997834719,786576828
       Cynthia Shields,5317142090,550153346
       Jackie Flores,1059336850,645808579
       Julie Moon,6341842976,737791474
       Bailey Patel,2717801668,438299022

/mnt/Forensics/evidence ❯ cat /mnt/Forensics/evidence/buyers_list.txt                                             21:43:48+0000
       File: /mnt/Forensics/evidence/buyers_list.txt

       Name,Handle,Darkweb Address
       Bugs Bunny,bugs123,https://bugs.onion
       Daffy Bunny,daffy123,https://daffy.onion
       Elmer Bunny,elmer123,https://elmer.onion
       Marvin Bunny,marvin123,https://marvin.onion
       Tweety Bunny,tweety123,https://tweety.onion

/mnt/Forensics/evidence ❯ cat /mnt/Forensics/evidence/cc_list.txt                                                 21:45:04+0000
       File: /mnt/Forensics/evidence/cc_list.txt

       Joseph Noble, 6502762836055918
       David Kirby, 30059985895796
       Amanda Jensen, 4192315835301278
       Robert Rodriguez, 349236707126751
       Mr. David Taylor, 4816298452554821503
       Cynthia Parks, 502046355173
       Leah Mcdaniel, 4013213521532826
       Michael Reyes, 341116823782869
       Stuart Price, 36534883361756
       Jerry Paul, 4148296999070716
```

Figure 33: Device 2 'bank_accounts.csv' 'buyers_list.txt' 'cc_list.txt'

Exhibit B – Financial Theft through Malicious Executables

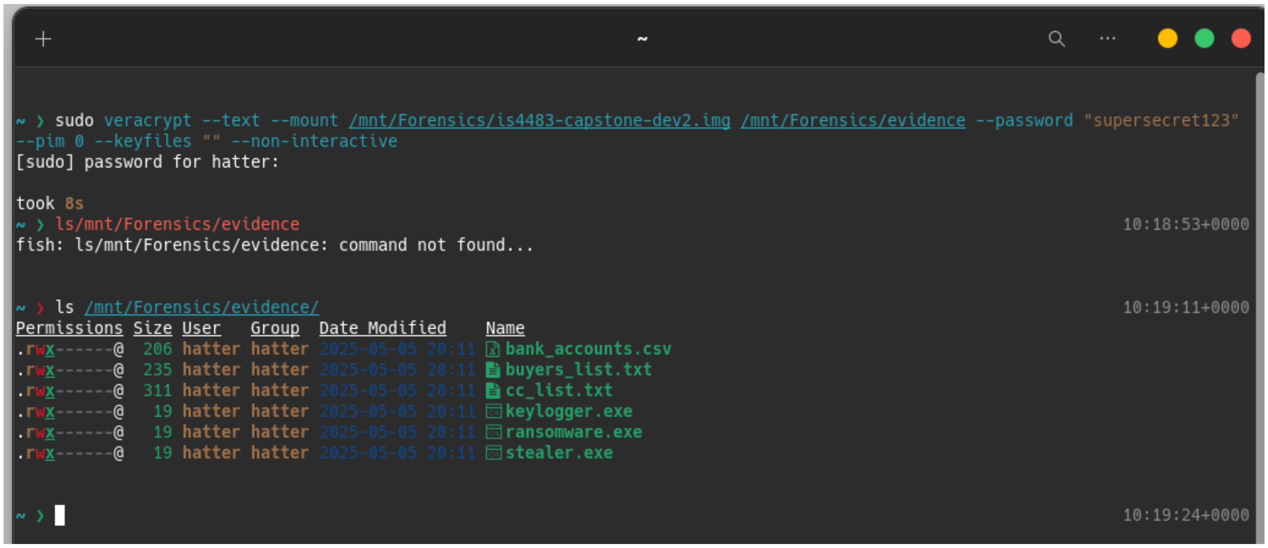


Figure 34: Device 2 contains malicious executables 'keylogger.exe' 'ransomware.exe' 'stealer.exe'

References

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DeepWiki. "Command Line Interface." *DeepWiki*, 4 Jan. 2026, deepwiki.com/veracrypt/VeraCrypt/2.3-command-line-interface. Accessed 13 May 2026.

openwall. "John/Run at Bleeding-Jumbo · Openwall/John." *GitHub*, 2025, github.com/openwall/john/tree/bleeding-jumbo/run. Accessed 13 May 2026.